A2_CP414_W2026

Saturday, February 21, 2026 7:32 PM

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CP414, Winter 2026 - Assignment 2

Upload **a single PDF file** to MyLS Dropbox. Your assignments in this class will be evaluated not only on the correctness of your answers, but on your ability to present your ideas clearly and logically.

1. Give regular expressions generating the following languages over the alphabet $\Sigma = \{0,1\}$:

a) L1 = the set of all strings that start with **1** or have odd length = $1(0\cup1)^* \cup (0\cup1)((00\cup01\cup10\cup11)^*$
b) L2 = the set of all strings that start with **0** and have even length = $(00\cup01)(00\cup01\cup11\cup11)^*$
c) L3 = the set of all strings that end with **1** and have even length = $(00\cup01\cup10\cup11)^*(01\cup11)$
d) L1 ∩ L2 = $\emptyset$
e) L2 ⋃ L3 = $((00\cup01)(00\cup01\cup10\cup11)^*) \cup ((00\cup01\cup10\cup11)^*(01\cup11))$
f) L2 ∩ L3 = $01\cup((00\cup01)(00\cup01\cup10\cup11)^*(01\cup11))$
g) The set of all strings that does not contain pattern **0010**.
h) The set of all strings except **111** and **01** = $\varepsilon\cup0\cup1\cup00\cup10\cup11\cup((000\cup001\cup010\cup011\cup100\cup101\cup110)(0\cup1)^*)$

2. Use the procedure described in Lemma 1.60 to convert the following NFAs to regular expressions:

(a) start state q_1, accepting state q_2

	0	1	ε
-> q_1	{ q_2}	{ q_1, q_2}	Ø
* q_2	{q_1}	Ø	Ø

(b) start state q_1, accepting state q_2

	0	1	ε
-> q_1	Ø	Ø	{q_3}
* q_2	{q_2, q_3}	{q_3}	Ø
q_3	{q_3}	{q_2}	{q_3}

Show your work (including GNFA and intermediate results after each state removal).

3. Convert the following regular expressions to NFAs using procedure given in Theorem 1.54. In all parts $\Sigma = \{0,1\}$.

a) $\mathbf{1(100)^* \cup 010}$

b) $\mathbf{1(10)^+ \cup (010)^*}$ = $1(10)(10)^* \cup (010)^*$

c) $\mathbf{(1\cup0^*)0^*1^+}$ = $(1\cup0^*)0^*11^*$

4 (a). Consider language $\mathbf{L = \{01^n01^n \mid n > 0\}}$ over alphabet $\Sigma=\{0,1\}$. Using Pumping Lemma prove that this language is not regular.

(b) Using result of **(a)** prove that language $\mathbf{B = \{ww \mid w \text{ from } \Sigma^*\}}$ is not regular. DO NOT USE PUMPING LEMMA! Use **closure** properties of regular languages instead.

5. Variation of Problem 1.53

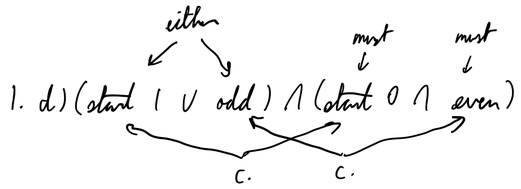
Let $\Sigma=\{0,1,2,3,\ldots,9,+,=\}$ and

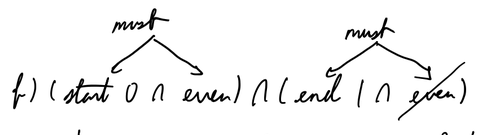
SUM = {a+b=c | a,b,c are unsigned integers, and c is the sum of a and b}.

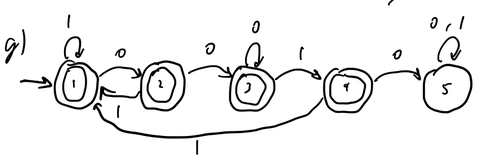
For example, string "**99+21=120**" is in **SUM**, while string "**99+21=77**" is not in **SUM**.

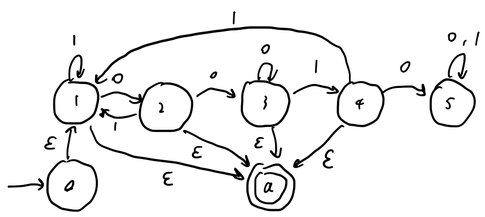
Prove that **SUM** is not a regular language.

Let p be pumping length, choose $s = a^p+b^p=c$.
$a^p+b^p=c \in SUM$, $|SUM|>2p>p$, s satisfy pumping lemma.
since $|xy| \le p = a$, y must be a digit of a.
pumping up: $a^{p+k}+b^p=c$, where a^k is a digit between 0 to 9, $k \ge 1$.
Since $a^p+b^p=c$, then $a^{p+k}+b^p > c$,
therefore $a^{p+k}+b^p=c \notin SUM$.
By contradiction, $SUM \notin R$.

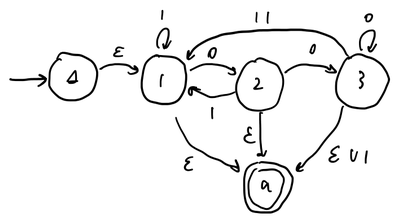




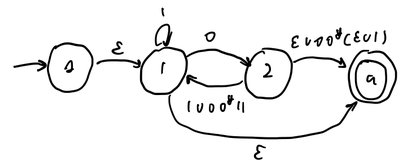




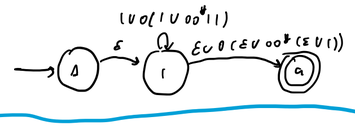
Remove 5:
ϕ
Remove 4:
$R_{31} = 11$
$R_{3a} = \varepsilon \cup 1$



Remove 3:
$R_{21} = 1 \cup 00^*11$
$R_{2a} = \varepsilon \cup 00^*(\varepsilon \cup 1)$

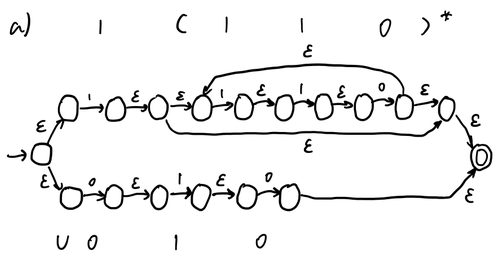


Remove 2:
$R_{1a} = \varepsilon \cup 0(\varepsilon \cup 00^*(\varepsilon \cup 1))$
$R_{11} = 1 \cup 0(1 \cup 00^*11)$

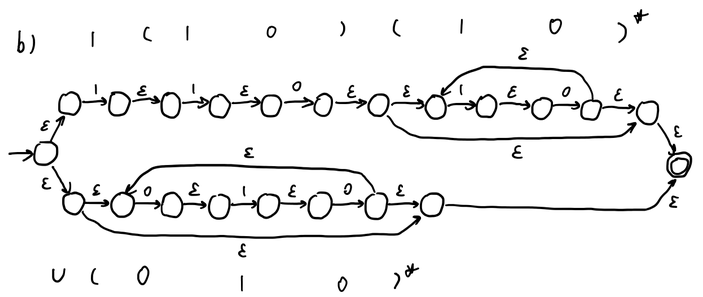


Remove 1:
$R_{sa} = (1 \cup 0(1 \cup 00^*11))^*$
$(\varepsilon \cup 0(\varepsilon \cup 00^*(\varepsilon \cup 1))$

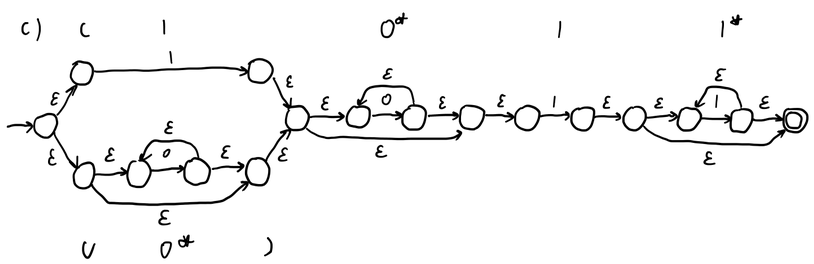
3. a)



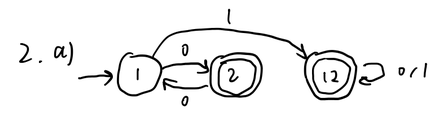
b)



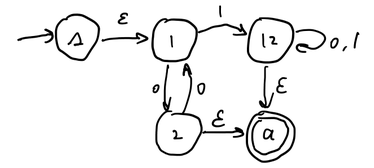
c)



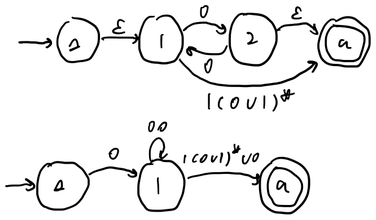
2. a)



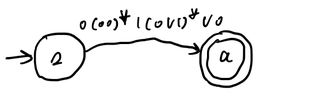
NFA to GNFA



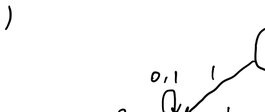
Remove 12:
$R_{1a} = 1(0\cup1)^*$



Remove 2:
$R_{1a} = 1(0\cup1)^*0 \cup 0$
$R_{11} = 00$

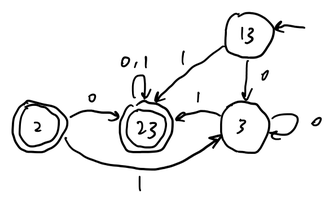


Remove 1:
$R_{sa} = 0(00)^*1(0\cup1)^*00$

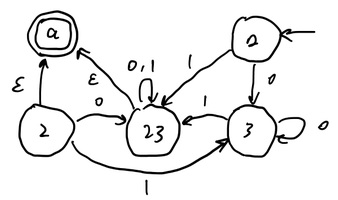


$0(00)^*1(0\cup1)^*00$

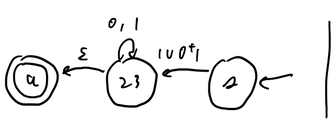
b)



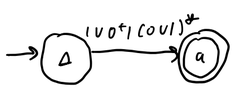
NFA to GNFA



Remove 2:
ϕ
Remove 3:
$R_{a,23} = 1\cup00^*1 = 1\cup0^+1$



Remove 23:
$R_{sa} = 1\cup0^+1(0\cup1)^*$



$1\cup0^+1(0\cup1)^*$

4. a) Let p be pumping length, choose $s = 01^p01^p$.
$01^p01^p \in L$ and $2p+2 > p$, s satisfies pumping lemma.
By splitting s to xyz, we get 3 cases.
Case 1: $y = 1^k$, $k \ge 1$. $x = 0$
Pumping up: $s = 01^{p+k}01^p \notin L$
Case 2: $y = 0^k$, $k \ge 1$. $x = \varepsilon$
Pumping up: $s = 0^k1^p01^p \notin L$
Case 3: $y = 01^k$, $k \ge 1$. $x = \varepsilon$
Pumping up: $s = 01^k01^{p+k}01^p \in L$
Since no case satisfy pumping lemma,
by contradiction, $L \notin R$.

b) Let $A = 01^*01^*$, such that $A \in R$,
By closure under intersection, $A \cap B \in R$.
$A \cap B = \{01^n01^n \mid n \ge 0\}$, where $w = 01^n$, $n \ge 0$.
From part a), it shows that $A \cap B \notin R$ for all $n > 0$.
By contradiction, B is not regular.